ModelIt!

Teacher's Guide

Phase Changes: When Matter Transforms

Same Stuff, Different State — and Nothing Is Lost

Quick Reference

Grade Level	Grade 05L2
NGSS Standard	5-PS1-2, 5-PS1-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 5-PS1-2, 5-PS1-4

Measure and graph quantities to provide evidence that regardless of the type of change that occurs when heating, cooling, or mixing substances, the total weight of matter is conserved. Conduct an investigation to determine whether the mixing of two or more substances results in new substances.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a model showing how temperature and pressure affect the distribution of matter across three states.
Disciplinary Core Idea	**PS1.A Structure and Properties of Matter** — Students model how matter changes state while total mass is conserved.
Crosscutting Concept	**Scale, Proportion, and Quantity** — Students observe that the total quantity of matter remains constant across phase changes.

CAST Testing Alignment

MULTIPLE CHOICE:

You weigh a sealed container of water, then boil it. The container now holds steam. How does the weight compare to before boiling?

CONSTRUCTED RESPONSE:

Explain what happens to water molecules as ice is heated to steam, and how you know that no matter is lost during this process.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Matter exists in three common states — solid, liquid, and gas. Phase changes occur when energy is added or removed, but the total amount of matter always stays the same. This is the law of conservation of matter.

Molecular Motion

In solids, molecules vibrate in fixed positions. In liquids, they slide past each other. In gases, they fly freely in all directions. Adding heat energy makes molecules move faster, causing phase changes.

Conservation of Mass

Antoine Lavoisier proved in the 1770s that matter is conserved in all changes. When water boils, it seems to disappear, but the same mass of water now exists as invisible steam.

Pressure Effects

Pressure affects phase changes. High pressure pushes gas molecules closer together, turning them back into liquids. This is how refrigerators and air conditioners work.

Phase Change Diagrams

Scientists use phase diagrams to show the temperatures and pressures where each state exists. Water's unique properties — like expanding when it freezes — make it essential for life.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify Temperature, Pressure, Solid Amount, Liquid Amount, and Gas Amount as the five system components.

E **Establish Relationships**
Students determine that Temperature reduces Solid Amount (-) and increases Gas Amount (+), while Pressure reduces Gas Amount (-) by compressing gas back to liquid.

V **Visualize & Model**
Students build the phase change model showing how matter transfers between three states while total mass remains constant.

E **Evaluate Outcomes**
Students run the Heating from Frozen scenario and observe Solid decreasing as Liquid and Gas increase, confirming conservation of matter.

R **Revise & Extend**
Students add Melting Point Threshold or Particle Speed to explore more precise phase change behavior.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Same stuff. Three forms. Does anything disappear when it changes?"*

Watch For

Hold up an ice cube and ask: Where will this water be in an hour?

Slide 2: Learning Objectives

2 min

Say: *"Today we prove that matter is NEVER lost — it just changes form."*

Watch For

Students read objectives and predict: Does boiled water weigh less?

Slide 3: Big Question

2 min

Say: *"Most people think water disappears when it boils. Let's test that."*

Watch For

Quick vote: Does water weigh less after boiling? Tally predictions.

Slide 4: LEVER Framework

2 min

Say: *"Our model tracks matter in three states at once."*

Watch For

Preview how the three internal components always add up to the same total.

Slide 5: Activity 1: Components

8 min

Say: *"Temperature and Pressure are external — we control them. The three states respond."*

Watch For

Students sort components and discuss which ones they can control in real life.

Slide 6: Activity 2: Connections

8 min

Say: *"Watch for the key pattern: what one state loses, another gains."*

Watch For

Guide students through the relationships, emphasizing conservation.

Slide 7: Activity 3: Simulation

10 min

Say: *"Slowly increase temperature and watch what happens to each state."*

Watch For

Students run Heating from Frozen scenario and track all three amounts.

Slide 8: Discoveries

5 min

Say: *"The total never changed. Matter does not disappear — it transforms."*

Watch For

Have students add all three amounts at each time step to verify total is constant.

Slide 9: STEM Challenge

10 min

Say: *"Now prove it with a real experiment. Design a test that shows boiled water does not disappear."*

Watch For

Groups design experiments; discuss how to capture steam.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Temperature, Pressure
- **INTERNAL:** Solid Amount, Liquid Amount, Gas Amount

Relationships

- **Temperature to Solid Amount:** NEGATIVE (-) — Increasing temperature causes solids to melt, reducing the amount of solid.
- **Temperature to Gas Amount:** POSITIVE (+) — Increasing temperature causes liquids to evaporate, increasing the amount of gas.
- **Pressure to Gas Amount:** NEGATIVE (-) — Increasing pressure compresses gas molecules back toward liquid form.
- **Temperature to Liquid Amount:** POSITIVE/NEGATIVE (+/-) — Moderate heat melts ice into liquid (+), but high heat evaporates liquid into gas (-). Liquid amount first rises, then falls.

Key Discoveries

- When temperature rises, solids melt into liquids and liquids evaporate into gases
- The TOTAL amount of matter stays the same — it just changes form
- High pressure can push gas molecules back into liquid form
- Solid + Liquid + Gas always equals the same total — matter is CONSERVED

Multiple Choice Post-Assessment Answer Key

1. In the phase change model, what happens when Temperature increases from 0% to 100%?

Correct: B) Solid Amount decreases, then Liquid Amount increases, then Gas Amount increases as matter changes state

Correct! As temperature rises, solid melts into liquid (Solid decreases, Liquid increases). At higher temperatures, liquid evaporates into gas (Liquid decreases, Gas increases). The matter transforms from one state to the next.

2. What is the most important pattern the model reveals about total mass during phase changes?

Correct: C) Solid Amount + Liquid Amount + Gas Amount always equals the same total, proving matter is conserved

Exactly! At every point in the simulation, the three amounts always add up to the same total. Matter is not created or destroyed during phase changes. It just moves from one state to another. This is the law of conservation of matter.

3. What type of relationship exists between Temperature and Gas Amount?

Correct: B) Positive: higher temperature increases gas as liquid evaporates into vapor

Correct! Higher temperatures give molecules more energy, causing them to break free from liquid form and become gas. More heat means more evaporation, which means more gas. This is a positive relationship.

4. You weigh a sealed container of water, then boil it so all the water becomes steam. How does the weight compare?

Correct: C) It weighs exactly the same because the same matter is inside, just in gas form

That's right! The sealed container holds the same molecules of water whether they are liquid or steam. No matter entered or left. The mass is identical because matter is conserved during phase changes.

5. In the model, what happens when Pressure is set to 90%?

Correct: B) Gas Amount decreases because high pressure compresses gas molecules back toward liquid form

Correct! High pressure squeezes gas molecules closer together, pushing them back into liquid form. This is a negative relationship between Pressure and Gas Amount. Refrigerators and air conditioners use this principle.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Water disappears when it boils"	Water does not disappear — it becomes water vapor (steam), an invisible gas. The same mass of water still exists, just in a different state.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Ice weighs less than water"	The same mass of water weighs the same whether it is ice, liquid, or steam. Ice floats because it is less DENSE (takes up more space), not because it weighs less.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Gas has no mass"	Gas absolutely has mass. A balloon inflated with air weighs more than an empty balloon. Gas molecules are just spread far apart.